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Strong-Lensing Time Delays in the Berger–Hopf Framework: An R1 Effective Calculation

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Abstract

In the Berger–Hopf framework, gravitational lensing is the incoming-to-outgoing transfer of environmental light through an envelope event. We investigate how its surface contributions can be represented in a conditional R1 effective calculation, keeping the microscopic common-source derivation distinct from the response already computed. A fixed closed background and cubic scalar–tensor action determine a regular finite-body metric response without a fitted local amplitude. Common incident light, measured redshifts, source coordinates and observer clocks define image directions and relative arrival times. For a $10^{11} M_{\odot}$, 3 kpc body at redshift 0.3 and a source at redshift 1, the screening radius is 212.0292 kpc. Relative to the specified fixed-body reference, the leading delay offsets are 25.9593 s and 8.6490 s. A positive conventional effective-mass profile reproduces the complete retained metric and optical response: the matched representations agree. A common finite-body general-relativistic second-order control changes the reference-relative offsets by less than 0.1 ms. Nuisance calculations identify interchangeable source and profile contributions. Explicit source applications and continuation equations advance the BHSM transfer calculation. Positive complement constructions show why microscopic matching needs additional physical data; material and outer-state prescriptions are required for further evolution.

Keywords: Berger–Hopf framework; strong gravitational lensing; time delays; envelope response; Vainshtein screening; effective-mass representation

1. Introduction

The Berger–Hopf Standard Model (BHSM) organizes this study around an envelope event traversed by environmental light. The physical object is the transfer from a prepared incoming signal to its outgoing image directions and observer arrival-time differences, conditional on the envelope state and surrounding geometry. We ask: how do BHSM surface contributions determine gravitational lensing and time delays for light traversing that event?

The governing premise is that conventional gravitating-matter and scalar descriptions can represent the same underlying surface contributions. The present calculation develops one conditional effective realization of that premise. Its closed cosmological reference branch, called R1, was specified in the Berger–Hopf research program before this local lens calculation [4]. The R1 action supplies normalized scalar and metric equations whose optical consequences can be calculated. We derive the cap/source pullback conditions and demonstrate that the leading R1 normalization permits inequivalent positive complements. A complete physical map from BHSM surface and boundary data to that action therefore

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consumes additional embedding and constitutive data; the adopted premise does not select those data.

Strong-lensing time delays compare light emitted by one variable source along different paths. Their interpretation depends on the local optical response, propagation geometry, source and environment. We therefore use one common optical preparation and state the admitted nuisance freedoms [1–3]. Photons follow metric null rays in the retained calculation; no additional photon interaction law is assumed. The cubic scalar–tensor action uses kinetic mixing, a mechanism also studied in the kinetic-braiding and Galileon literature [5–7].

The construction proceeds from the prepared envelope and environment to the action, local source and boundary conditions, then to the metric, optical encounter, images and delays. We retain the earlier source-free target-state family, whose additional response vanishes in its stated boundary problem, and calculate a matter-sourced response of the distinct rolling scalar. Its normalization is fixed by the declared R1 action and background, without a fitted local amplitude. A positive conventional effective mass subsequently reproduces the entire retained scalar optical field. This constructive equivalence is a result of the BHSM lensing program: a distinctive departure from conventional gravity is not a condition for the calculation's success.

The density profile is illustrative and no observed lens is fitted. Neither the R1 branch nor its local effective action is established as a unique microscopic completion of the Berger–Hopf framework. Numerical results concern the stated classical action and approximation; reference-relative offsets are not unavoidable discrepancies between fully matched conventional and scalar representations.

2. Envelope State and Environmental Preparation

The incoming marker probes the event; it does not select the envelope reference state. The conditional state consists of the frozen R1 branch, the finite local density excess and the regular local boundary branch specified below. The earlier source-free target state is retained as a separate null control in Appendix A. Homogeneous geometry, local matter and the local scalar response are accounted for once.

2.1. Incident light, finite body and environment

The lens and source redshifts are $z_l = 0.3$ and $z_s = 1$. The incoming test light is a narrow transient marker with identical emission phase, spectrum, polarization and physical transverse source coordinate in the effective response and its reference. The observed reference wavelength is 550 nm. The fixed source displacement is $s = (0.0014576633674174603, 0)$ Mpc. The light intensity is taken in the test-field limit, so its gravitational backreaction is neglected. The observables are image directions and observer proper-time differences between copies of that marker.

The local body in the conditional envelope is prescribed as a finite excess density above the homogeneous environment. The effective response and fixed-body reference use this same profile:

$$\rho(r) = \frac{15M}{8\pi R^3}\left(1 - \frac{r^2}{R^2}\right) \quad (0 \le r \le R), \qquad M = 10^{11} M_\odot, \quad R = 3\ \text{kpc}. \tag{1}$$

It vanishes outside the body. This is the quadratic-density Tolman–VII profile [8], used as a controlled spherical mass distribution. Define

$$m_g = \frac{GM}{c^2}, \qquad f_M(r) = \frac{M(<r)}{M} = \begin{cases} \frac{1}{2}(5x^3 - 3x^5), & x = r/R \le 1, \\ 1, & r \ge R. \end{cases} \tag{2}$$

For the effective scalar response, the density is prescribed and the leading pressure adjustment is calculated separately. We do not reuse the exact Einstein-fluid pressure as an exact modified-gravity solution.

The physical Jacobi maps are independently initialized for observer–lens, observer–source and lens–source propagation. They share one background, the same transported screen basis, and frequency normalization. At the reference state their scalar values are

$$(D_{ol}, D_{os}, D_{ls}) = (945.162839373,\ 1689.312239687,\ 1076.994348060)\ \text{Mpc}. \tag{3}$$

The main body is excluded from these maps and enters the local encounter once. The local response calculation holds the maps fixed, isolating the encounter contribution. Its fixed-body reference uses the same R1 geometry; it is not a fit of a separate complete ΛCDM cosmology.

3. Action, Normalization, Local Source and Boundary Data

3.1. Inherited surface source and the scope of its reduction

The inherited BHSM source organization assigns the surface and gravitational contributions to explicit slots:

$$\begin{aligned} \mathcal{L}_\psi \psi &= J_{\text{grav}} + \mathcal{E}_\Sigma[Q_\Sigma] + \xi, \\ Q_\Sigma &= \frac{c^2}{8\pi} \int_{\Sigma_\star} H_\Sigma \Delta K_{uu}\, \mathrm{d}A, \\ h_{\text{topo}} &= -\mathcal{A}_g^{-1} \mathcal{B}_{g\psi} \psi. \end{aligned} \tag{4}$$

These are the retained source and constrained metric-response definitions, not a newly closed constitutive law. The surface functional uses the seam curvature data; its export operator, orientation and full action normalization remain to be matched to the local field variables. Here ψ names that inherited topographic variable, not the independent rolling scalar T, the target field η, or the metric coordinate ψ_{geo} introduced below. The local metric Hessian in this source organization is also distinct from the cosmological ADM constraint block. No stochastic forcing or population covariance is required for the specified deterministic encounter. The term ξ remains a source slot; it is not supplied with an invented realization or state law.

Table 1. Status of the source-to-observable construction.

Status	Statement
Adopted BHSM premise	Environmental light traverses an envelope event; conventional and scalar descriptions represent its surface contributions.
Derived effective result	The declared R1 action fixes the local matter source, regular scalar branch and metric response, hence the optical transfer.
Derived equivalence	A positive effective mass reproduces that retained metric and optical field.
Derived source condition	Cap momentum and Schur forcing specify the required pullback; positive same-domain lifts prove its microscopic completion is not unique from the leading coefficients.
Open common-source map	Reduction of the BHSM surface and boundary data to the normalized R1 source, with the same independent variables and validity range.

The calculation below executes the effective problem and its optical application. The derived common-source conditions specify the physical selection data consumed by a further application within the BHSM program.

3.2. Effective action and frozen R1 normalization

We use signature $(-,+,+,+)$ and $M_{\rm Pl}^{-2} = 8\pi G$. In the action and local field equations, $c = \hbar = 1$. The conditional R1 representation uses the retained scalar–tensor action

$$S = \int \sqrt{-g}\,\mathrm{d}^4x \left[\frac{M_{\rm Pl}^2}{2} R - \frac{1}{2}(\nabla T)^2 + \frac{(\nabla T)^2 \Box T}{\lambda} - V_0 + \mu^3 T + \mathcal{L}_m \right], \qquad \lambda = \Lambda^3 > 0. \quad (5)$$

The field T is an independent scalar. It is distinct from the target field η and from a metric conformal coordinate. For example, $\gamma_{ij} = a^2 e^{2\psi_{\rm geo}} \bar{\gamma}_{ij}$ gives $\psi_{\rm geo} = -\Psi$ to linear order in Newtonian gauge; it does not supply another independent scalar in addition to the metric.

The dimensionless background convention is $\widehat{T} = T/M_{\rm Pl}$, $\widehat{v} = \dot{\widehat{T}}/(M_{\rm Pl}H_0)$, $\widehat{d} = \ddot{\widehat{T}}/(M_{\rm Pl}H_0^2)$, $\widehat{\lambda} = \lambda/(M_{\rm Pl}H_0^2)$. The frozen parameters are

$$\begin{aligned} (\Omega_{m0}, \Omega_{r0}, \Omega_{k0}) &= (0.31, 9\times10^{-5}, -0.018), \\ H_0 &= 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \widehat{\lambda} = 1, \qquad \frac{\mu^3}{M_{\rm Pl}H_0^2} = 3, \\ \frac{V_0}{M_{\rm Pl}^2 H_0^2} &= 2.3986073449708067. \end{aligned} \quad (6)$$

They define one reference branch, not a framework-wide calibration. The stored background solution is reused without refitting. The reference spatial curvature is $K = -\Omega_{k0}H_0^2/c^2 > 0$. Its unperturbed angular-diameter maps obey $D_{ab} = a_b \sin[\sqrt{K}(\chi_b - \chi_a)]/\sqrt{K}$. The previously computed six-amplitude sensitivity of one normalized axial $n = 2$ harmonic is retained as a separate propagation control; its amplitudes are zero in the reference maps used for the local comparison. Its homogeneous scalar current,

$$J = \dot{\bar{T}} + \frac{6H\dot{\bar{T}}^2}{\lambda}, \qquad \dot{J} + 3HJ = \mu^3, \quad (7)$$

also provides a sign and normalization check on the local variation.

3.3. Reduction from the retained effective action

Varying Equation (5) gives

$$\Box T + \frac{2}{\lambda}\left[(\nabla_\mu \nabla_\nu T)^2 - (\Box T)^2 + R_{\mu\nu}\nabla^\mu T \nabla^\nu T \right] - V_T = 0. \quad (8)$$

The Ricci term couples the scalar response to matter through the same Einstein equation. No independent phenomenological matter coupling is introduced. Write $T = \bar{T}(t) + \pi_T(r)$ and freeze $v = \dot{\bar{T}}$, $d = \ddot{\bar{T}}$ and H at the lens epoch. After subtracting the homogeneous equations, retain leading weak metric potentials and quasi-static derivatives, including the nonlinear spatial Hessian responsible for screening. Set

$$b_T = \frac{2v^2}{\lambda}, \qquad C = 1 + \frac{4(d + 2Hv)}{\lambda} - \frac{2v^4}{M_{\rm Pl}^2\lambda^2}. \quad (9)$$

The leading common constraints are

$$2M_{\rm Pl}^2 \Delta\Phi = \delta\rho - b_T \Delta\pi_T, \qquad \Phi = \Psi, \quad (10)$$

$$C\Delta\pi_T - \frac{2}{\lambda}\left[(\Delta\pi_T)^2 - (\partial_i\partial_j\pi_T)^2 \right] = -\frac{b_T}{2M_{\rm Pl}^2}\delta\rho. \quad (11)$$

The last term of C follows from eliminating the metric constraint. The equal potentials are a result at the retained order; they are not an exact assertion about a fully nonlinear space-time.

At $z_l = 0.3$ the frozen solution yields

$$\begin{aligned} H/H_0 &= 1.1836105772, & \widehat{v} &= 0.2377644264, \\ \widehat{d} &= 0.2484715891, & b_T/M_{\rm Pl} &= 0.1130638449, \\ C &= 4.2388585594. \end{aligned} \tag{12}$$

The physical b_T has mass dimension one. In numerical units $\widehat{\lambda} = 1$ is fixed; it is not a new local adjustable length.

3.4. Regular finite-body branch

For spherical symmetry the scalar equation integrates to

$$Cr^2\pi_T' - \frac{4}{\lambda} r(\pi_T')^2 = -b_T GM(<r). \tag{13}$$

The origin is regular and carries no additional scalar charge. The root continuously connected to the decaying weak-source solution is

$$\pi_T' = -\frac{2b_T GM(<r)}{r^2\left[C + \sqrt{C^2 + 16 b_T GM(<r)/(\lambda r^3)}\right]}. \tag{14}$$

It has $\pi_T' \propto r$ at the center and no interface delta-function source at R.

3.5. Executed source and material-boundary application

The source consumed by this reduction can be made explicit. Define an additional metric energy source ρ_Σ and independent scalar Euler source J_T by the pre-elimination rows $2M_{\rm Pl}^2\Delta\Phi = \rho + \rho_\Sigma - b_T\Delta\pi_T$ and $A\Delta\pi_T - 2[(\Delta\pi_T)^2 - |\,\mathrm{Hess}\,\pi_T|^2]/\lambda + b_T\Delta\Phi = J_T$, where $A = C + b_T^2/(2M_{\rm Pl}^2)$. The actual reduced source and spherical flux are

$$\begin{aligned} J_{\rm eff} &= J_T - \frac{b_T}{2M_{\rm Pl}^2}(\rho + \rho_\Sigma), \\ Cr^2\pi_T' - \frac{4}{\lambda} r(\pi_T')^2 &= \int_0^r s^2 J_{\rm eff}(s)\,{\rm d}s. \end{aligned} \tag{15}$$

For the declared action there is no extra layer term: $\rho_\Sigma = J_T = 0$, while the volume source remains nonzero. For a proper-radial thin layer with coefficients σ_Σ and $j_{T,\Sigma}$, its retained weak-metric jump would instead be

$$[Cr^2\pi_T' - 4r(\pi_T')^2/\lambda]_{R^-}^{R^+} = R^2\left(j_{T,\Sigma} - \frac{b_T\sigma_\Sigma}{2M_{\rm Pl}^2}\right). \tag{16}$$

These are source applications of the original coupled equations; a surface charge is not an independently adjustable lensing amplitude. An additional anisotropic interface stress would also consume the corresponding metric slip equation.

Using the frozen coefficients, the computed surface section is $\pi_T'(R)/(M_{\rm Pl}H_0) = -2.119825778845613 \times 10^{-4}$. In $M_{\rm Pl} = H_0 = c = 1$ units, with radius in c/H_0, an independent density integration gives $\int_0^R r^2 J_{\rm eff}\,{\rm d}r = -1.216417408185084 \times 10^{-13}$. The maximum relative nonlinear-flux residual at $r/R = 0.1, 0.5, 1, 2, 10$ is 4.15×10^{-16}; the original integrated metric Gauss residual is zero at printed precision. Independent interior quadratic and exterior rational roots give a relative material-boundary flux mismatch 6.23×10^{-16}.

The separate exact-GR material control has continuous same-oriented K_{uu} and hence zero material Q_Σ. That material worldtube is distinct from the cosmological cap seam, and its zero layer does not remove the nonzero volume source. The receipt is `artifacts/strong_lensing_bhsm_integration_20261020/source_application.json`. Appendix E derives the cap/source pullback and shows constructively why the leading normalization does not select a unique microscopic completion.

4. Metric and Optical Encounter Response

The common metric constraint gives

$$\Phi' = \Psi' = \frac{GM(<r)}{r^2} - \frac{b_T}{2M_{\rm Pl}^2}\pi_T', \qquad F(r) \equiv \frac{\delta\Phi'}{\Phi_N'} = \frac{2\epsilon_\infty}{1+\sqrt{1+r_V^3 f_M(r)/r^3}}, \tag{17}$$

where

$$\epsilon_\infty = \frac{b_T^2}{2M_{\rm Pl}^2 C}, \qquad r_V^3 = \frac{16 b_T GM}{\lambda C^2}. \tag{18}$$

The last expression uses $c = 1$. The numerical conversion is

$$r_V^3 = 16(b_T/M_{\rm Pl})m_g(c/H_0)^2/(\widehat{\lambda}C^2).$$

Deep in the exterior screened region $F(r)$ approaches $2\epsilon_\infty(r/r_V)^{3/2}$; at large radius it approaches ϵ_∞. The potential itself still decays. This is the usual cubic screening mechanism [6,7], evaluated here with the frozen normalization and a regular extended body.

The corresponding leading hydrostatic correction is

$$\delta p(r) = \int_r^R \rho(s)\,\delta\Phi'(s)\,{\rm d}s, \qquad \delta p(R) = 0. \tag{19}$$

It is positive and changes the central Newtonian pressure by 3.56959×10^{-6} fractionally. This supports the specified leading-order matter configuration; it does not construct an exact modified Tolman–Oppenheimer–Volkoff sequence or prove radial stability. Holding density fixed and changing its required pressure does not impose an independently fixed common material equation of state on both models.

4.1. Projection into the optical encounter

Restore c and interpret Φ as the dimensionless metric potential. For physical impact radius $b_{\rm imp}$ the additional deflection is

$$\delta\hat{\alpha}(b_{\rm imp}) = 4b_{\rm imp}\int_0^\infty \frac{\delta\Phi'(\sqrt{b_{\rm imp}^2+z^2})}{\sqrt{b_{\rm imp}^2+z^2}}{\rm d}z, \tag{20}$$

$$\delta\hat{\psi}(b_{\rm imp}) = \int_R^{b_{\rm imp}} \delta\hat{\alpha}(s)\,{\rm d}s. \tag{21}$$

The arbitrary zero of potential does not affect relative delays. We calculate the gradient and Hessian consistently from this same projected potential. A local force ratio is not substituted for a projected photon deflection ratio.

The leading stationary delay change between image i and reference image 0 is

$$\delta\Delta t_{i0} = -\frac{1+z_l}{c}\left[\delta\hat{\psi}(b_i) - \delta\hat{\psi}(b_0)\right]. \tag{22}$$

The finite result also resolves the image positions in Equation (23). The projection freezes the local coefficients during the encounter and extends the isolated local solution through the thin plane. Matching to the linear perturbation is assumed in an overlap region $r_V \ll r \ll c/H_l$; a global closed-cosmology solution has not been constructed.

5. Image Directions and Observer Delays

5.1. Image directions and observer arrival-time differences

Let $x = D_{ol}\theta$, $\beta = D_{os}^{-1}s$ and $\hat{\alpha} = \nabla_x\hat{\psi}$, where $\hat{\psi}$ has length units. The common thin-lens equations are

$$\beta = \theta - D_{os}^{-1}D_{ls}\hat{\alpha}(D_{ol}\theta), \tag{23}$$

$$ct(\theta) = \frac{1}{2}(\theta-\beta)^T\mathsf{T}(\theta-\beta) - (1+z_l)\hat{\psi}(D_{ol}\theta) + f(\beta),$$

$$\mathsf{T} = (1+z_l)D_{ol}^T D_{ls}^{-1}D_{os}. \tag{24}$$

The maps must satisfy reciprocity so that T is symmetric [1]. A common $f(\beta)$ cancels in relative delays. Image positions are resolved again after changing the encounter.

For smooth geometric variations, fixing s implies $\delta\beta = -D_{os}^{-1}\delta D_{os}\beta$; fixed physical and fixed angular source prescriptions cannot be imposed simultaneously. At fixed measured redshift, the stationary first variation is

$$c\delta t = \frac{1}{2}q^T\delta\mathsf{T}q - q^T\mathsf{T}\delta\beta - (1+z_l)\left[\hat{\alpha}^T\delta D_{ol}\theta + \delta\hat{\psi}|_x\right] + \delta f, \qquad q = \theta - \beta. \tag{25}$$

The image-position variation cancels at a stationary image. In the local R1 comparison $\delta D_{ab} = \delta\beta = 0$ and the nonzero intrinsic term is retained. Appendix B records the measured-redshift convention for comparisons that vary the maps.

5.2. Evaluated response and reference-relative offsets

The screened branch predicts $r_V = 212.0292$ kpc and $\epsilon_\infty = 0.001507886$, or a 0.1507886% asymptotic force enhancement. At $r = R = 3$ kpc the local force enhancement is 5.06706 parts per million (ppm), whereas the deflection enhancement at the same impact radius is 11.7467 ppm. The ray samples the outer, less screened field as well as the field near closest approach.

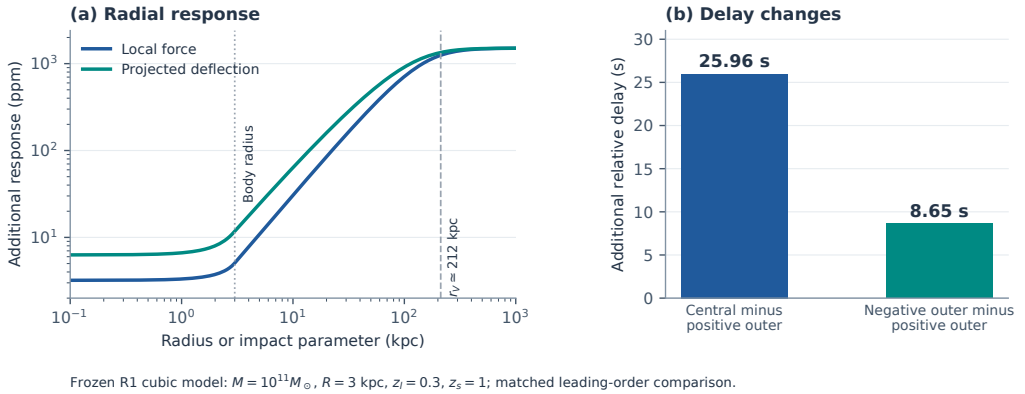
Table 2. Leading-order image positions for the effective response and its fixed-body reference. Image 0 is the positive outer image, image 1 the central image, and image 2 the negative outer image. Printed digits specify reproducible model values, not absolute astrometric accuracy.

Image	Reference (arcsec)	R1 response (arcsec)	Offset (μas)
0	+0.835476034	+0.835482157	+6.1233
1	−0.082161940	−0.082161152	+0.7881
2	−0.657495525	−0.657499874	−4.3486

Table 3. Relative delays for the same source marker, with image 0 as reference. The offsets are relative to the specified fixed-body reference, with the physical light preparation held fixed.

Observable	Fixed-body reference	R1 response	Offset
Δt_{10} (days)	21.716076566	21.716377020	25.9593 s
Δt_{20} (days)	14.271452387	14.271552491	8.6490 s
$\Delta t_{10}/\Delta t_{20}$	1.521644467	1.521654847	6.8212 ppm

Figure 1 compares the radial response and the two delay changes. The nonzero response follows from the rolling scalar already in Equation (5). It does not alter the earlier constant-target null result in Appendix A. An independent singular isothermal control, with velocity dispersion 220 km s^{-1} and the same reference source, retains images at $+1.067883$ and -0.711922 arcsec and a 17.094532-day delay. That analytic control validates the optical conventions; the finite body in Tables 2 and 3 is the local boundary problem used for the evaluated response.



Frozen R1 cubic model: $M = 10^{11} M_\odot$, $R = 3$ kpc, $z_l = 0.3$, $z_s = 1$; matched leading-order comparison.

Figure 1. R1 envelope response relative to the fixed-body reference. Left: additional local force and projected deflection as fractions of their fixed-body reference values. The distinction reflects integration along the ray. Right: reference-relative offsets in the two delays for the fixed source preparation. Neither panel is an observational detection forecast.

At fixed epoch, $r_V \propto M^{1/3}$. The same frozen background also specifies the redshift dependence, without refitting:

Lens redshift	ϵ_∞	r_V (kpc), $M = 10^{11} M_\odot$
0.0	0.00351132	248.300
0.3	0.00150789	212.029
0.5	0.00086052	191.230
1.0	0.00022832	150.778
1.5	0.00007085	123.011

This is a prediction for the coefficients and force law, not a delay table for identical angular sources at these different redshifts.

The supplied reference-run outputs give the following implementation checks. The integrated scalar flux has maximum relative residual 7.9×10^{-16}. A direct line-of-sight quadrature and an independent angular-substitution quadrature agree on the additional deflection to 4.5×10^{-13} relative. The image equation residual is below 8.5×10^{-22} rad. Differentiating the projected potential and deflection independently gives relative discrepancies below 2.0×10^{-9} and 2.4×10^{-12}, respectively. Finite re-solved and stationary first-order delay changes differ by 0.00016 s and 0.000056 s. These are numerical and perturbative checks within the model, not bounds on omitted physics. In the current Windows replay, the corresponding flux residual is 8.30×10^{-16} and the deflection-derivative discrepancy is 3.19×10^{-12}; the remaining differences are recorded with their numerical tolerances in the replay receipt. The supplied and replay outputs remain separate records.

6. Conventional Representation and Consistency Checks

6.1. Positive effective-mass representation of the same response

The retained metric and optical response admit an explicit conventional representation. Define an effective additional enclosed mass by

$$\delta M_{\rm eff}(r) = M f_M(r) F(r), \qquad \delta\rho_{\rm eff}(r) = \frac{M}{4\pi r^2}\frac{\mathrm{d}}{\mathrm{d}r}[f_M(r)F(r)]. \tag{26}$$

This density is nonnegative. To see this, put $S = \sqrt{1 + r_V^3 f_M / r^3}$ and $n_M = \mathrm{d}\ln f_M / \mathrm{d}\ln r \geq 0$. Then

$$\frac{\mathrm{d}\ln(f_M F)}{\mathrm{d}\ln r} = \frac{1}{2}\Big[(1+S^{-1})n_M + 3(1-S^{-1})\Big] \geq 0. \tag{27}$$

The ordinary Poisson field of $M(<r) + \delta M_{\rm eff}(r)$ has exactly the same $\Phi = \Psi$ as Equation (17) within the retained approximation. It therefore reproduces the full deflection field, relative potential, images and delays, including multiple source planes in the same geometry. Its asymptotic extra mass is

$$\delta M_{\rm eff}(\infty) = \epsilon_\infty M = 1.50788624 \times 10^8 M_\odot. \tag{28}$$

The rank-one multiple-source residual discussed below distinguishes the scalar from the restricted Tolman–VII parameter family, not from all conventional matter distributions. Because slow matter and light respond to the same retained potentials, ordinary dynamical data do not automatically break this equivalence either. Independent restrictions on the matter/environment model or an observable outside this retained equivalence would be needed to distinguish those representations. No such independent likelihood is asserted here, and such a distinction is not required for the envelope-transfer calculation.

The effective mass in Equation (26) represents the same retained contribution as the scalar correction in Equation (17). One uses either the explicit scalar correction or its effective-mass representation in the encounter, not their sum. The construction proves equivalence of the retained metric and optical response. It does not yet derive both descriptions from one normalized microscopic BHSM surface source. That common-source derivation remains open without changing the framework organizing the calculation.

6.2. The ordinary finite-body control

For the conventional control, a regular Einstein-fluid interior can be matched to Schwarzschild. Its compactness is $m_g/R = 1.59514 \times 10^{-6}$. The leading optical application above uses the weak-field projection of this density.

6.3. A common second-order ordinary-gravity control

To test the effect of the first omitted conventional deflection, we expand the exact static GR optical scattering integral of the same Tolman–VII body through second post-Minkowskian order (2PM). Put $\varepsilon_g = m_g/R$, $y = b_{\rm imp}/R$ and $t = \sqrt{1-y^2}$. The additional radial deflection is

$$\delta\hat{\alpha}_{\rm GR2}(b_{\rm imp}) = 2\varepsilon_g^2 I(y), \tag{29}$$

$$I(y) = \begin{cases} \dfrac{15(\arcsin y - yt)}{4y^2} + y\left(\dfrac{15}{2}t - 4t^5 - \dfrac{48}{7}t^7\right), & 0 < y < 1, \\ \dfrac{15\pi}{8y^2}, & y \geq 1. \end{cases} \tag{30}$$

At the center $I(0) = 0$ and $I'(0) = -6/7$; the interior coefficient includes the leading fluid pressure and the conversion from areal radius to invariant asymptotic impact. At and

outside the surface it matches the standard $15\pi m_g^2/(4b_{\rm imp}^2)$ exterior deflection [9]. The consistent plane potential is

$$\delta\hat{\psi}_{\rm GR2}(b_{\rm imp}) = 2\varepsilon_g^2 R J(y), \qquad J'(y) = I(y), \qquad J(1) = 0. \tag{31}$$

The supplied implementation evaluates I, I' and J analytically.

The R1 response and fixed-body reference receive the identical GR2 potential, gradient and Hessian, and both image systems are re-solved. The R1 coefficients and leading scalar profile are held fixed. The scattering impact is identified with the physical thin-plane impact in this control. This checks robustness to the common GR term; it does not compute the missing scalar–GR higher-order terms or a complete finite-distance 2PM cosmological arrival functional. The replay command is

```
python scripts/run_strong_lensing_gr2_control.py
```

6.4. Result of the common GR correction

At the original two exterior-image impacts, the GR2 term is 3.68 and 4.68 ppm of the leading conventional deflection. Its inclusion changes the conventional image coordinates by $(+1.35456, -0.01519, -1.72123)$ microarcsec and the conventional relative delays by $(+8.28473, +2.55245)$ s. The absolute correction is therefore material at the scale of the calculated reference-relative image shifts.

The response-minus-fixed-body-reference offsets are considerably more stable under this common-term check.

Table 4. Robustness of the reference-relative R1 offsets under the same GR2 correction in both calculations. The final column is the change in the reference-relative offset, not the absolute GR correction.

Response offset	Leading calculation	Common GR2 added	Change
$\delta\Delta t_{10}$ (s)	25.95925815	25.95932714	$+68.99$ μs
$\delta\Delta t_{20}$ (s)	8.64901938	8.64900924	-10.14 μs
$\delta\theta_0$ (μas)	$+6.12327625$	$+6.12326934$	-6.91×10^{-6} μas
$\delta\theta_1$ (μas)	$+0.78810716$	$+0.78810711$	-5.58×10^{-8} μas
$\delta\theta_2$ (μas)	-4.34859153	-4.34858643	$+5.09 \times 10^{-6}$ μas

The two relative changes in the reference-relative delay offsets are $+2.66$ and -1.17 ppm. The delay-ratio response remains 6.8212 ppm. Thus the nonzero reference-relative response survives this specific common-term control without changing its quoted precision in seconds or microarcseconds.

An independent mixed stationary-action derivative reproduces the change in the reference-relative delay offsets within 0.7 ns; this checks the small drift without subtracting four re-solved delays.

The analytic correction was also compared with direct null-geodesic scattering in the solved static GR fluid and exterior metric. At the three reference impacts, the exact deflection minus the 1PM and 2PM terms has magnitude below 1.8×10^{-16} rad for this body's compactness $\varepsilon_g = 1.59514 \times 10^{-6}$. A compactness extrapolation independently recovers the interior 2PM coefficient, and direct integration verifies $J' = I$. The latter checks establish the implemented ordinary-GR scattering coefficient, not a physical error bound for the scalar model or its cosmological embedding. The full corrected coordinates, delays and checks are supplied in `gr2_control.json` and the review outputs.

6.5. Single-source constraints and nuisance freedoms

For an observable residual $\delta y = y - y_0$, write

$$\delta y = B\,\delta a + N\,\delta \nu + \epsilon, \qquad W\mathsf{C}W^T = I, \qquad P = I - (WN)(WN)^{+}. \tag{32}$$

Here B is a physical response, N the stated nuisance response, C a specified covariance, and $+$ a Moore–Penrose inverse. The rank of PWB concerns distinguishability in that nuisance model. A zero direction is unconstrained by the corresponding likelihood; a Fisher pseudoinverse does not turn it into a zero error bar. We take C positive definite. For the screened comparison, the single column is a fixed reference-relative response template; rank one denotes a surviving residual direction, not identification of a freely varied fundamental scalar parameter.

Using source position, body mass and radius, and a compensated mass-sheet scale as nuisances, the single R1-response column has projected rank one if the body center is fixed. Adding both center coordinates gives rank zero. The raw delay shifts in Table 3 are therefore not nuisance-independent observables for this configuration. The rank calculation uses 0.001 arcsec and 0.1 day as numerical whitening scales, not as demonstrated observational precisions.

The ordinary mass-sheet transformation,

$$\psi_\lambda(\theta) = \lambda\psi(\theta) + \frac{1-\lambda}{2}|\theta|^2, \qquad \beta_\lambda = \lambda\beta, \qquad \Delta t_\lambda = \lambda\Delta t, \tag{33}$$

cancels from the ratio of two delays. It does not remove source, centering, radial-profile, or independent line-of-sight freedoms [3]. Consequently the 6.8212 ppm ratio change is a fixed-preparation prediction, not a unique diagnostic of a distinct representation.

For the separately computed axial smooth-map response, the images of a softened isothermal model depend only on $\beta = s/D_{os}$, $\theta_E = b_{\rm phys} D_{ls}/D_{os}$, and $\theta_c = r_c/D_{ol}$, while all delays share the factor $(1+z_l)D_{ol}D_{os}/(cD_{ls})$. Varying the three scalar maps can therefore be reproduced by these source/profile changes and a delay scale. The saved six-anchor response accordingly has zero projected rank in the unrestricted nuisance model. This analytic factorization concerns that family; it is not a universal theorem that cosmological lensing information always vanishes.

6.6. A bounded multiple-source test

A supplemental design uses one body and three sources at $z_s = (0.8, 1.2, 1.8)$, with source offsets $0.1\theta_{E,s}$ at angles $(0.15, 1.1, 2.2)$ rad. These choices were fixed before applying the screened response. The nuisance model includes independent source positions and delay scales, a shared mass, radius and center, and shared physical convergence and two shear components. The resulting scalar column retains rank one; adding independent source-plane convergence columns leaves rank one as well.

This surviving component is small: its norm is 0.1202% of the raw whitened response in the expanded nuisance model. In the same illustrative weighting, the residual coordinates reach approximately 6 nanoarcsec and the residual delays 0.086 s. The explicit 24 observable weights and response arrays are supplied with the code. These tidal nuisance fields are phenomenological controls. This test does not solve the nonlinear scalar equation for a nonspherical galaxy and its environment, and it is not a signal-to-noise forecast. Its rank is specific to the finite radial profile family.

7. Computational Methods and Reproducibility

The calculation uses the fixed manuscript source revision `49a1dab` and records full input hashes. The main reproduction command from the repository root is

```
python scripts/run_strong_lensing_r1_screened.py
```

The coefficient, independent projection, and multiple-source checks are under `scripts/r1_screened_review/`. Numerical results and their units are under `artifacts/strong_lensing_r1_screened_20261020/`. The research package additions and exact commands are integrated with the pinned repository background and optical dependencies. The recorded package-era replay of three portable review scripts reproduced all 305 saved numerical values in that environment. The current integration replayed nine numerical commands, compared 1161 saved numerical values, and passed 16 focused checks without changing the original numerical artifacts. These are separate execution records, not independent physical measurements. The reproducibility manifest identifies versions and hashes rather than equating successful execution with physical validation.

OpenAI ChatGPT and Codex materially assisted derivation, code development, numerical checks, interpretation and manuscript revision. The independent quadrature uses the same physical radial equation but a separately implemented integration variable and root calculation. Its agreement is a reproducibility check, not independent empirical evidence or external human peer review.

8. Discussion: BHSM Interpretation, Accuracy and Remaining Derivation

The calculation supplies a nonzero local response with an amplitude fixed by the stated action and background. The known cubic screening mechanism is not itself new. The specific contribution is its evaluated finite-body envelope transfer in this reference branch, a consistent common incident-light comparison, and a constructive conventional representation of the retained response. The adopted BHSM premise places those descriptions within the same surface-contribution program; the calculation establishes their retained optical equivalence while leaving the normalized microscopic common-source map unfinished.

The numerical precision must be separated from physical accuracy. The evaluated response and fixed-body reference use leading weak-field, thin-lens equations. Sections 6.3 and 6.4 explicitly address the common ordinary-GR second-order term. Its effect on the absolute image coordinates reaches 1.72 microarcsec, but its effect on the offset from the fixed-body reference is below 7×10^{-6} microarcsec. The two delay differences change by less than 0.1 ms. This successful control does not establish absolute microarcsecond astrometric accuracy or an equivalent physical time-delay error bound. The scalar–GR mixed terms, scalar corrections to the higher-order fluid geometry, and finite-distance cosmological 2PM arrival terms remain uncalculated. The exact effective-halo equivalence in Section 6 is established at the retained leading scalar order; this common-term test neither extends that proof to all orders nor changes its retained-order scope.

The scale separation $H_l r_V/c = 5.64 \times 10^{-5}$ supports a local quasi-static treatment. In the formal isolated projection the contribution to the extra deflection beyond 20 Mpc is about 1.1–1.7 $\times 10^{-6}$ of that extra deflection. Neither check bounds environmental, finite-distance, global matching, or omitted-sector errors. The reference is spherical; external matter, time dependence, a finite source, plasma propagation and microlensing would require additional physical modeling before comparison with data. The central image is included in the mathematical census; its detectability is not assumed. For a nonzero environmental scalar, the cubic equation also contains mixed local/environmental Hessian terms. Adding an optical tide alone does not solve that modified local scalar boundary problem.

The leading scalar principal coefficients are positive on the retained regular branch, as detailed in Appendix C. However, the radial scalar characteristic speed squared reaches approximately 1.474 in units of the metric light speed squared. This is a property of the scalar effective theory, while the photons used here propagate on metric null rays. Positive leading coefficients do not prove complete stability or a causal ultraviolet completion.

The pure cubic specialization does not establish the normalization or irrelevance of an additional short-distance topographic operator. A low-momentum expansion with a spatial fourth-order term cannot be extrapolated to galaxy scales without its coefficient and validity range. Independent cap/scalar reactions, the general map from exported topographic forcing into a target phase, and global closed-cosmology matching remain open. They do not authorize an arbitrary local lensing amplitude. The constant-target result and the present rolling-scalar calculation remain distinct conditional sectors.

Appendix D derives the exact bulk current balance, its rolling-reference obstruction, the environmental cross-Hessian and the material/outer-domain conditions consumed by the next application. These equations do not constitute an evolved higher-order solution.

The next BHSM derivation is therefore specific: reduce the surface, cap and topographic boundary contributions in the same physical variables to a normalized source and operator for the independent R1 scalar and its metric constraints. The reduction must specify the boundary/state prescription and the complement spectrum or cutoff that controls any higher-derivative expansion. It must also connect the local response to the surrounding closed geometry. Neither identifying the rolling scalar with the target phase or metric coordinate, nor adding an adjustable optical amplitude, supplies that relation. The present effective calculation is the evaluated transfer problem to which that derivation must connect.

9. Conclusions

For the prescribed $10^{11} M_{\odot}$, 3 kpc body at $z_l = 0.3$ and the common source at $z_s = 1$, the conditional cubic R1 envelope calculation gives offsets in the relative delays of approximately 25.96 s and 8.65 s, image changes of approximately 0.79–6.12 microarcsec, and a 6.82 ppm change in the delay ratio. The screening radius is approximately 212 kpc. These are reproducible offsets from the fixed-body reference in the declared leading-order calculation, obtained without fitting a new scalar amplitude. Adding the same finite-body GR2 correction to both calculations changes the delay differences by only $+69$ and -10 microseconds in the stated thin-plane control.

The linearized single-source response is absorbed by the admitted source, mass, radius, mass-sheet and center freedoms. A three-source finite-profile comparison retains a small shape response, but a positive conventional effective halo reproduces the entire retained scalar optical field. Thus prepared light measures the envelope transfer, while the stated nuisance model controls which properties can be assigned separately. The positive effective-mass construction is a constructive result: fully matched conventional and scalar representations share the demonstrated leading optical response. The source application and positive-complement construction determine the consumed source relation and prove that its microscopic completion is not selected by the leading coefficients. A physical cap-to-T embedding and the specified material/global state are needed to place this effective calculation within a complete common-source construction.

Supplementary Materials: The integrated research package contains the local-action derivation, numerical outputs, independent projection and nuisance checks, the common GR2 control, source manifests, and exact reproduction commands. The historical transport and target-sector derivations remain in the pinned repository.

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Conflicts of Interest: The author declares no conflicts of interest.

Appendix A. The Constant-Target Null Control

The earlier source-free target sector has

$$S_\eta = -\int N\sqrt{h}\, w_0 F(X)\, \mathrm{d}t\, \mathrm{d}^3x, \qquad F(X) = \frac{\kappa_\eta}{2}X + \frac{1}{8}X^4, \qquad X = g^{\mu\nu}\langle D_\mu\eta, D_\nu\eta\rangle, \quad \text{(A1)}$$

with $\eta = (s,z) \in S^6 \subset \mathbb{R}\oplus\mathbb{C}^3$, $s^2 + 2z^\dagger z = 1$, target metric $\mathrm{d}s^2 + 2\,\mathrm{Re}(\mathrm{d}z^\dagger \mathrm{d}z)$, and $w_0\kappa_\eta > 0$. For $\eta_q = (\cos q, \sin q\, e_1/\sqrt{2})$ with constant q and zero connection, $D\eta = 0$, $F(0) = 0$, and the target current and stress vanish. For a radial meridian $q(r)$, regularity sets the conserved flux

$$r^2 e^\nu \sqrt{f_r}\, w_0(\kappa_\eta + X^3)q' = 0, \qquad X = f_r(q')^2 \quad \text{(A2)}$$

to zero. Positivity gives $q' = 0$. This proves the null only for the stated source-free regular boundary problem.

A constant background still permits positive perturbation stiffness. For a nonconstant Dirichlet phase perturbation, the quadratic energy and its second variation (phase self-curvature) are

$$E_\eta^{(2)}[\varphi] = \tfrac{1}{2}w_0\kappa_\eta \int e^{\Phi-\Psi}|\nabla\varphi|^2\, \mathrm{d}^3x,$$

$$\delta^2 E_\eta[\varphi,\varphi] = 2E_\eta^{(2)}[\varphi] > 0.$$

The mixed metric/phase response vanishes at $D\eta = 0$. The solved matter body therefore has ordinary nonzero lensing and zero additional response in this target family. A vanishing matched seam contribution is not a statement that all possible gravitational or boundary sources vanish. The rolling scalar T in Equation (5) is a different explicitly specified field.

Appendix B. Moving Measured-Redshift Endpoints

For an observer-normalized affine ray, $J_a'' = \mathcal{T} J_a$ with $(J_a, J_a')_a = (0, \omega_a I)$. Let $Z_{\lambda,a} = \delta \ln \omega|_{\lambda_a}$ and $Z_{\chi,a} = \delta \ln \omega|_{\chi_a}$. On the reference ray,

$$\begin{aligned} Z_{\chi,a} &= Z_{\lambda,a} - \mathcal{H}_a \delta\chi_{\mathrm{ray},a}|_\lambda, \\ \delta\lambda_a &= -\frac{Z_{\lambda,a}}{(\ln \overline{\omega})'_{\lambda,a}}, \qquad \delta\chi_a = \delta\chi_{\mathrm{ray},a}|_\lambda + \overline{\chi}'_{\lambda,a}\delta\lambda_a = -\frac{Z_{\chi,a}}{\mathcal{H}_a}. \end{aligned} \tag{A3}$$

Thus converting an affine shift alone does not give the full radial event shift. The moving-vertex initial conditions are

$$\delta J_a|_a = -\omega_a I\, \delta\lambda_a, \qquad \delta J_a'|_a = \Delta\omega_a I, \qquad \Delta\omega_a = \delta\omega_a|_\lambda + \overline{\omega}'_a \delta\lambda_a. \tag{A4}$$

At fixed measured redshift and observer normalization, $\Delta\omega_a = 0$. Output event and frame conversions must also be included. These conventions are needed for the three compatible maps; they are inactive in the local comparison where the maps are held fixed.

Appendix C. Leading Scalar Principal Coefficients

After eliminating the leading metric mixing, the homogeneous time coefficient is

$$D = 1 + \frac{12Hv}{\lambda} + \frac{6v^4}{M_{\mathrm{Pl}}^2 \lambda^2} = 4.3962210287. \tag{A5}$$

For $\mathcal{S} = \sqrt{C^2 + 16 b_T GM(<r)/(\lambda r^3)}$, the leading local diagonal coefficients are

$$K_r = \mathcal{S}, \qquad K_\Omega = \mathcal{S} + \frac{1}{2} r\mathcal{S}', \qquad K_t = D + \frac{3}{2}(\mathcal{S} - C) + \frac{1}{2} r\mathcal{S}'. \tag{A6}$$

Nonnegative density implies $K_\Omega \geq (\mathcal{S}^2 + 3C^2)/(4\mathcal{S}) > 0$. For this branch K_t is also positive; the frozen homogeneous ratio is $C/D = 0.9642051$. The exterior maximum of K_r/K_t is 1.47366 at approximately 100.107 kpc. These are leading local coefficients. Mixed time–space and higher weak-field terms, the full coupled spectrum, and an ultraviolet completion are outside this check.

Appendix D. Exact Bulk Continuation, Environment and Matching

The retained action also specifies the next field problem. The field equations below use $c = \hbar = 1$, restoring c in scale comparisons. Its exact bulk equations can be derived without supplying the unfinished microscopic surface-to-R1 map. They are distinct from an evolved physical solution with a specified material law and global state. The full spherical component derivation is supplied in `docs/strong_lensing_r1_screened_20261020/R1_REMAINING_FIELD_DERIVATION_20261020.md`.

Appendix D.1. The current balance and the rolling reference

Put $U_\mu = \nabla_\mu T$, $Y = U_\mu U^\mu$, $\Pi_{\mu\nu} = \nabla_\mu \nabla_\nu T$, $S = \Box T$, $\mathcal{K}_\Pi = \Pi_{\mu\nu} U^\mu U^\nu$ and $g_3 = 2/\lambda$. The exact metric equation and cubic stress are

$$\begin{aligned} M_{\mathrm{Pl}}^2 G_{\mu\nu} &= T^m_{\mu\nu} + U_\mu U_\nu - g_{\mu\nu}(Y/2 + V) + T^{(3)}_{\mu\nu}, \\ T^{(3)}_{\mu\nu} &= g_3[-S U_\mu U_\nu - g_{\mu\nu}\mathcal{K}_\Pi + 2U_{(\mu}\Pi_{\nu)\alpha}U^\alpha]. \end{aligned} \tag{A7}$$

Equation (8) has the exact current form

$$J^\mu = U^\mu + g_3(\Pi^{\mu\nu}U_\nu - S U^\mu), \qquad \nabla_\mu J^\mu = -\mu^3. \tag{A8}$$

On the homogeneous reference $J^0 = -J$ for the positive current in Equation (7), reproducing its source sign. For a spherical patch $ds^2 = -N^2dt^2 + A^2dr^2 + r^2d\Omega^2$, the actual balance is

$$\partial_t(NAr^2J^t) + \partial_r(NAr^2J^r) = -\mu^3 NAr^2. \tag{A9}$$

An exactly static metric with $T = vt + \pi(r)$ and constant v would remove the temporal term. A regular center would then require

$$J^r(r) = -\frac{\mu^3}{NAr^2}\int_0^r N(s)A(s)s^2\,\mathrm{d}s, \tag{A10}$$

which grows as $-\mu^3 r/3$ in a static asymptotically flat region, rather than giving the decaying isolated excess. Moreover $V_0 - \mu^3 vt - \mu^3\pi$ makes the stress time dependent for nonzero $\mu^3 v$. The rolling background time-current balance must therefore be retained before its homogeneous contribution is subtracted. The executed quasi-static solution does this at its retained order; appending a static GR2 potential does not solve this full continuation. Small $H_l r_V/c$ supports the local freezing approximation but is not a physical delay-error bound.

Appendix D.2. The environmental contraction

At the retained quasi-static order let $\pi_T = \pi_L + \pi_E$ and $\delta\rho = \rho_L + \rho_E$. Subtracting the environmental equation from the total scalar equation gives the actual local application

$$\begin{aligned} C\Delta\pi_L - \frac{2}{\lambda}[(\Delta\pi_L)^2 - \Pi_L : \Pi_L] \\ - \frac{4}{\lambda}[(\Delta\pi_E)(\Delta\pi_L) - \Pi_E : \Pi_L] = -\frac{b_T\rho_L}{2M_{\mathrm{Pl}}^2}, \end{aligned} \tag{A11}$$

where $\Pi_{L,ij} = \partial_i\partial_j\pi_L$ and similarly for E. The environmental input is the full spatial Hessian in $[(\mathrm{tr}\,\Pi_E)\delta^{ij} - \Pi_E^{ij}]\partial_i\partial_j\pi_L$, not its trace alone. Constant or uniform-gradient scalar data have no such cross term, whereas a tidal Hessian generally does. A projected optical tide does not determine this three-dimensional contraction. Physical environmental density and boundary/Cauchy data must be solved with the same scalar and metric constraints; their total principal operator must also be checked. The optical nuisance tides in the main text do not perform that solve.

Appendix D.3. Material and outer-domain defining data

An evolved perfect fluid consumes a material relation, for example $\rho_m = \rho_m(n)$ and $p_m = n\,\mathrm{d}\rho_m/\mathrm{d}n - \rho_m$, together with $\nabla_\mu T_m^{\mu\nu} = 0$. At static equilibrium the radial application is $p_m' = -(\rho_m + p_m)\nu'$ for $N = e^\nu$. A common fixed barotropic law with $\delta\rho_m = 0$ would give $\delta p_m = 0$, whereas Equation (19) gives a positive pressure change. The retained comparison therefore prescribes common density, not a common fixed material equation of state. A physical material continuation must allow density readjustment, specify additional support, or explicitly retain the prescribed-stress comparison.

In an overlap worldtube $r_V \ll r_m \ll c/H_l$, the regular branch fixes its far-zone derivative data without a new local charge:

$$\begin{aligned} \pi_T'(r_m) &= -\frac{b_T GM}{Cr_m^2}[1 + O((r_V/r_m)^3)], \\ \Phi'(r_m) = \Psi'(r_m) &= \frac{GM}{r_m^2}[1 + \epsilon_\infty + O((r_V/r_m)^3)]. \end{aligned} \tag{A12}$$

Smooth matching also requires the induced metric and extrinsic curvature, the scalar value and normal derivative, and the current balance in Equation (A9). Scalar additive data and

clock normalization belong to the outer state. A physical surface source would instead require its variational junction conditions; no moving-seam law is assumed here.

The flat local constraint cannot be extended unchanged to the entire compact slice. Its formal compact integral would require $\int \delta\rho \, \mathrm{d}V = 0$, which the isolated positive excess does not satisfy. The actual outer problem must retain curvature, time dependence and the homogeneous response, or a declared compensating excess. This is a compatibility check of the naive extension, not a prohibition of positive local mass in a closed cosmology. The outer solution must match both the local field and the source/observer clock protocol. Thus the exact bulk and environmental equations are available, while their material and global state data must be specified before a further physical solution is claimed.

Appendix E. Common-Source Pullback and a Nonuniqueness Construction

The original cap reduction is a constrained critical value with retained boundary traces and kernel coordinates. Its actual matcher variation defines the oriented canonical momenta by $\delta S^{\text{on-shell}}_{5,\epsilon} = \int \Pi^{ab}_{\epsilon} \delta\gamma_{\epsilon ab}$ and $S_{\text{match}} = \sum_{\epsilon} \int \Lambda^{ab}_{\epsilon}(h_{ab} - \gamma_{\epsilon ab})$. It gives

$$\Pi^{ab}_{\epsilon} = \Lambda^{ab}_{\epsilon}, \qquad 2C_{\partial} G^{ab}[h] = T^{ab}_{m} + \tau^{ab}_{\text{cap}}, \qquad \tau^{ab}_{\text{cap}} = \frac{2}{\sqrt{-h}} \sum_{\epsilon} \Pi^{ab}_{\epsilon}. \tag{A13}$$

Thus $u_a u_b \delta\tau^{ab}_{\text{cap}}$ supplies the metric energy source once a cap solution and observer are specified. The cap scalar normal momentum and the interface Euler variation supply different source slots; their pullback into canonical T must be established. The cap sum cannot be replaced with the reflection-even mean without its factor of two. A cap response absorbed into the local action is accounted for once.

For original coupled rows $A_P\psi + Bq = J_P$, $B^{\dagger}\psi + D_Q q = J_Q$, elimination on an invertible owned complement gives both the operator and its forcing:

$$(A_P - BD^{-1}_{Q,\perp} B^{\dagger})\psi = J_P - BD^{-1}_{Q,\perp} J_Q - Bq_0, \qquad q_0 \in \ker D_Q. \tag{A14}$$

Kernel compatibility equations remain explicit. A reduced Hessian alone does not determine this source. The physical embedding of T, its kinetic normalization, cubic vertex and source pairing must agree under the same reduction.

The leading R1 coefficients do not select a unique complement. To prove this, take two mathematical spectral lifts on the same ball $L = 10\pi\ell_0$, both real fields regular at zero and Dirichlet at L. The normalized radial modes are $f_n = \sin(n\pi r/L)/(r\sqrt{2\pi L})$, $n \geq 1$, with $q = p\ell_0^2 = (n/10)^2$. Define

$$H(q) = \begin{pmatrix} (C+1)q & g\sqrt{q} \\ g\sqrt{q} & M_c^2 + q \end{pmatrix}, \quad K_{\text{time}} = \text{diag}(D, 1), \quad (M_c^2, g) = (1,1) \text{ or } (4,2). \tag{A15}$$

Here M_c, g, q are dimensionless; D is the temporal coefficient in Appendix C. The same retained source is used and the complement is explicitly unforced for this counterexample. Neither lift is selected as a microscopic galaxy state. Each complement is positive, and its reduced kernel is $Cq + g^2q^2/[M_c^2(M_c^2+q)] > 0$ for every owned $q > 0$. Both leading coefficients are exactly C, D, yet after normalizing by C their fourth-order coefficients are $B/\ell_0^2 = 1/C = 0.2359125661$ and $1/(4C) = 0.05897814152$, with expansion scales $p\ell_0^2 \ll 1$ and $p\ell_0^2 \ll 4$, respectively. The common unit ℓ_0 has not been assigned a physical galaxy scale. This constructive nonuniqueness result identifies missing physical selection data rather than inferring them from an unavailable array.

A further direct identification can be excluded. The inherited distributed seam law $\nabla \cdot \mathbf{F}_{\mathrm{top}} = a\Theta(r - r_\star)/r^2$ with $F^r_{\mathrm{top}}(r_\star) = 0$ integrates to $F^r_{\mathrm{top}} = a(1/r - r_\star/r^2)$. Its growing integrated charge cannot be identified by a constant normalization with the isolated R1 flux, which decays as r^{-2}. A different finite-domain export or nonlocal source map remains possible. The full action/source trail, including the interface Euler relation, is supplied in `docs/strong_lensing_r1_screened_20261020/BHSM_COMMON_SOURCE_DERIVATION_20261020.md`. It specifies the physical embedding, constitutive interface density and boundary prescription needed for a unique further application.

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